

PHASE 3 • MODAL ANALYSIS, DUAL OBSERVABILITY & TRI-HYBRID CONTROL

Linear State Space → Modal Controllability → Dual Observability (4 vs 11 Transducers) → Tri-Hybrid Synthesis → Sensor-in-the-Loop Benchmark

Sylvester Subspace Solver

Real Ordered Schur CARE

Tracking Anti-Windup Clamping

1. INPUT FROM PHASE 2 & MODAL MAP

A ∈ ℝ^{11×11}, B ∈ ℝ^{11×4}

A. Continuous-Time Plant State-Space:

δdx/dt = A δx + B δu + B_d δd, y = C δx

B. Open-Loop Eigenvalues & Time Constants:

λ_M = −0.020 h^{−1} (τ ≈ 50h • Substrate drainage)

λ_T = −0.150 h^{−1} (τ ≈ 6.7h • Greenhouse thermal)

λ_L = −0.500 h^{−1} (τ ≈ 2.0h • Illumination)

λ_H = 0.000 h^{−1} (Integrator mode • Fan/Mist balance)

System Stability: Marginally stable due to one integrator mode (λ_H=0). All 10 remaining modes are strictly in the open left-half s-plane.

2. CONTROLLABILITY PROOF

Kalman Staircase Form

A. Kalman Controllability Matrix:

Co = [B, AB, A²B, ..., A¹⁰B] ∈ ℝ^{11 × 44}

rank(Co) = 4 of 11

B. Controllability Decomposition Proof:

T^T A T = [A_{cc} (4×4), A₁₂; 0, A_{uu} (7×7)]

T^T B = [B_{cc} (4×4); 0 (7×4)]

Stabilizability Guarantee: Uncontrollable substrate modes (NPK, pH, EC, tank) have negative real parts (λ ≤ −0.02) • The system is strictly **stabilizable**.

3. DUAL OBSERVABILITY PROOF

Case A vs Case B Sensor Suite

Case A: 4 Primary Microclimate Sensors

C_{prim} = [I_{4×4}, 0_{4×7}] ⇒ rank(Ob) = 4 / 11

Proof: Unobservable subspace has dimension 7. Soil nutrients (N, P, K), pH, EC, and water level cannot be detected from air data!

Case B: Dedicated 11-Transducer Suite

C_{full} = I_{11×11} ⇒ rank(Ob) = 11 / 11 (Full Rank!)

Proof: All 11 internal biological and reservoir states are 100% reconstructible.

Viva Core Concept: Mathematically proves why low-cost greenhouses fail when omitting root-zone and nutrient transducers!

4. DECENTRALIZED MULTI-LOOP PID

Tracking Anti-Windup Clamping

A. 4 Single-Input Single-Output Loops:

Loop 1: Moisture (y₁) → Irrigation Pump (u₁)

Loop 2: Temperature (y₂) → Ventilation Fan (u₂)

Loop 3: Humidity (y₃) → Ultrasonic Misting (u₃)

Loop 4: Illumination (y₄) → Supplemental Light (u₄)

B. Tracking Anti-Windup Clamping Logic:

IF (u_{raw} ≥ 1.0 ∧ e > 0) ∨ (u_{raw} ≤ 0.0 ∧ e < 0) THEN
 f_e(k+1) = f_e(k) *(Freeze integrator accumulation)*
ELSE f_e(k+1) = f_e(k) + e(k)·dt

Prevents massive actuator overshoots and long recovery delays caused by physical actuator saturation boundaries!

5. SCALED SUBSPACE POLE PLACEMENT

Sylvester Subspace Projection

A. Target Closed-Loop Pole Locations:

λ_{des} = {−0.60, −0.80, −1.00, −1.20} h^{−1}

Settling times: 1.0 – 2.0 hours (Safe physical actuation).

B. Sylvester Equation Subspace Solver:

(A_{env} − λ_i I) x_j = B_{env} g_j ⇒ K_{env} = G X^{−1}

Iterative optimization selects G to minimize cond(X).

Gain Explosion Elimination: Classical pole placement yields gains > 10⁷. Subspace scaling bounds max gain to **|K_{ij}| ≤ 0.2167** (< 1.0).

6. OPTIMAL LQR WITH INTEGRAL ACTION

Hamiltonian Real Ordered Schur CARE

A. Augmented 15-State System:

x_a = [x; z] ∈ ℝ¹⁵, dz/dt = y_{meas}(1:4) − r_{sp}

A_a = [A, 0; C_{track}, 0], B_a = [B; 0]

B. Symplectic Real Ordered Schur CARE Solver:

H = [A_a, −B_aR^{−1}B_a^T; −Q_a, −A_a^T] ∈ ℝ^{30×30}

[U, T] = schur(H, 'real'); ordschur(U, T, 'lhp')

P = Re(V₂ V₁^{−1}) ⇒ K_a = R^{−1} B_a^T P = [K_x, K_i]

Algebraic Precision: CARE residual norm = **1.25 × 10^{−10}**. Guarantees asymptotic zero steady-state error & 89.7% error variance reduction!

7. TRANSDUCER LAYER & SAFETY INTERLOCK

ESP32 Hardware Simulation

A. Physical Sensor Transduction & Noise:

y_{meas,i} = clamp(x_i + v_i, min_i, max_i), v_i ~ N(0, σ_i²)

Enforces 12-bit ADC quantization and sensor clipping ranges.

B. Emergency Water Safety Cutoff Interlock:

IF x₁₀ < 10.0% THEN u_{actual}(1) = 0.0 (Pump Cutoff)

C. Natural Thermodynamic State Bounding:

0 ≤ M ≤ 100%, 0 ≤ RH ≤ 100%, 0 ≤ W ≤ 100%, L ≥ 0

Hardware Protection: Eliminates pump cavitation, seal destruction, and unrealistic negative water/humidity numerical artifacts.

8. PERFORMANCE SCORECARD & SUMMARY

Quantitative Controller Comparison

Metric	Decentralized PID	Pole Placement	Optimal LQR-I
Total ISE	1.53 × 10 ⁸	8.04 × 10 ⁸	1.57 × 10⁷ (−89.7%)
Total IAE	50,307.27	118,518.36	5,854.26
Total Variation (TV)	2,090.93	277.93	728.28
Mean Crop Score (CCI)	99.76%	98.37%	99.94%
Max Feedback Gain	0.2000	0.2167	1.6540

Comprehensive Synthesis:

- LQR-I** delivers the tightest multivariable tracking & noise rejection.
- Pole Placement** minimizes actuator chattering (TV=277.93).
- Decentralized PID** is simplest but suffers from cross-coupling chattering.

1. Linear Plant

A (11x11), B (11x4)

2. Controllability

rank(Co) = 4/11

3. Observability

4/11 vs 11/11 Dedicated

4. Controller Design

PID / PP / LQR-I CARE

5. Sensor & Safety

Transducer Noise & Cutoff

6. 24h Diurnal Sim

5 Growth Stages

7. Crop Scorecard

CCI 99.94% (LQR-I)